

Optimizing Pre-Release Promotional Spend via Predictive Popularity Modeling

8,582	2,547	4,869	52 / 99
TOTAL TRACKS	UNIQUE ARTISTS	UNIQUE ALBUMS	MEAN POPULARITY

1. Background

Streaming platforms now onboard thousands of new tracks daily, but promotional budgets, editorial playlist slots, and A&R attention are finite and have to be committed before a track has any meaningful streaming history behind it. For labels, artist managers, and platform curation teams, the ability to read a track's commercial potential from what's known at release — not after the fact — directly affects marketing ROI and catalog strategy.

2. Problem Statement

Track popularity in this catalog is highly uneven — scores span 0 to 99 with an average of just 52, meaning a large share of releases never gain real traction. Decision-makers currently lean on artist reputation and instinct to decide where to place promotional effort, which doesn't scale as release volume grows and tends to reinforce already-established artists over new ones. What's missing is a systematic way to read the metadata available at or near release time — artist standing, followers, album context, track positioning, duration, explicit flag — and translate it into a defensible signal for how a track is likely to perform, before streaming data exists to confirm or deny it. Without this, promotional spend and playlist real estate keep getting allocated reactively instead of proactively.

3. Objective

Estimate a track's likely popularity from pre-release metadata alone, giving marketing and A&R teams a ranked, data-backed view of where to concentrate promotional spend — cutting wasted investment on low-potential releases and surfacing high-potential tracks that artist reputation alone would overlook.

4. Data Science Angle

This is primarily a regression problem where `track_popularity` is a continuous 0-99 score, and the goal is to estimate that value from features like `artist_popularity`, `artist_followers`, `album_type`,

`track_duration_min` , and `explicit` , rather than sorting tracks into fixed buckets. The same model can be recast as classification later — flagging tracks above a business-agreed threshold as "high-potential" — once stakeholders define what that cutoff should be. Either way, an EDA/correlation pass has to come first, since it determines which features are actually worth feeding into the model.

5. Key Business Questions

- Which artist- and album-level attributes predict track popularity most strongly, and is that relationship reliable enough to act on?
- Can pre-release metadata alone predict popularity well enough to guide promotional decisions, or is behavioral/streaming data a hard requirement?
- Do "underrated" artists exist in the catalog — high followers, historically low track popularity — representing under-promoted upside rather than genuinely low demand?
- Does explicit content, track duration, or album type systematically shift popularity, and should that inform release strategy?
- How much of a track's popularity is really artist-brand halo versus the track itself — and does that change how new/emerging artists should be evaluated?
- Where should the "high-potential" cutoff sit for a classification use case, and what's the tradeoff between wasted promo spend (false positives) and missed hits (false negatives)?